

GILLETTE NORTHEAST WYOMING, WY, USA

WMO: 726650

Lat: 44.339N

Lon: 105.542W

Elev: 4354

StdP: 12.53

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94023

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.3	-2.0	-15.2	2.8	-4.3	-8.8	4.0	-1.2	33.9	39.7	30.8	37.2	6.8	340	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.5	95.0	61.9	91.6	61.1	88.7	61.0	66.3	84.0	64.9	82.3	63.6	81.1	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	95.1	70.8	59.4	88.9	70.5	57.3	82.5	70.4	33.4	83.4	32.2	82.2	31.1	80.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	26.1	23.4	DB	-16.0	100.3	7.3	3.2	-21.2	102.5	-25.5	104.4	-29.5	106.1	-34.8	108.4	(4)
(5)			WB	-16.4	69.0	7.1	1.2	-21.6	69.9	-25.8	70.6	-29.8	71.3	-35.0	72.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	26.1	25.9	35.8	43.0	51.8	62.9	72.3	70.3	60.3	46.0	34.9	26.1
	DBStd	19.32	11.83	13.21	11.97	9.63	8.71	7.90	6.19	6.68	9.51	9.92	11.95	12.18
	HDD50	3623	740	675	454	249	84	5	0	0	23	190	463	740
	HDD65	7347	1205	1094	906	661	414	130	12	30	196	591	903	1205
	CDD50	2312	0	1	13	38	141	393	690	628	332	66	10	0
	CDD65	563	0	0	0	0	6	69	237	193	56	2	0	0
	CDH74	8231	0	0	3	26	159	1099	3254	2643	985	61	1	0
	CDH80	3761	0	0	0	1	32	449	1629	1248	391	11	0	0
Wind	WSAvg	10.5	11.3	11.3	11.3	11.5	10.7	9.7	9.0	9.2	9.6	10.5	11.2	11.3
Precipitation	PrecAvg	16.5	0.6	0.6	0.7	1.9	3.0	3.3	1.5	1.3	1.3	1.2	0.7	0.6
	PrecMax	22.3	1.3	1.5	1.6	4.0	11.1	10.0	3.4	4.0	2.7	3.1	2.1	1.3
	PrecMin	12.0	0.1	0.1	0.2	0.3	0.2	0.1	0.5	0.0	0.0	0.0	0.2	0.1
	PrecStd	2.6	0.3	0.4	0.4	1.1	2.3	2.1	0.9	1.2	0.8	0.8	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	60.8	71.7	77.7	84.1	96.5	100.2	97.8	92.9	81.2	70.4	57.0
		MCWB	40.5	44.1	48.2	52.4	58.3	61.5	63.2	62.8	59.8	54.2	47.7	42.8
	2%	DB	48.8	54.0	64.5	72.1	78.7	89.8	96.5	93.7	89.2	74.3	63.0	51.5
		MCWB	38.0	40.8	45.7	50.7	57.0	61.4	62.7	60.8	58.7	51.2	44.8	39.5
	5%	DB	45.4	47.9	59.6	66.1	73.5	85.0	92.7	90.8	84.5	69.1	57.4	46.3
		MCWB	36.2	38.0	43.8	48.6	56.0	61.3	62.1	60.7	57.0	49.7	42.3	36.6
	10%	DB	42.4	43.2	54.2	60.9	69.4	80.7	89.5	87.7	80.2	63.4	52.6	43.1
		MCWB	34.5	34.8	41.4	46.0	53.8	60.6	62.2	60.4	55.9	47.6	40.2	34.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	44.8	49.9	54.6	62.1	67.8	68.5	67.2	64.9	56.1	48.4	43.0
		MCDB	52.9	59.2	68.5	73.7	76.7	82.1	86.7	87.7	81.0	74.8	67.4	55.9
	2%	WB	38.7	41.3	46.5	51.4	59.3	65.2	66.7	65.3	61.1	53.0	45.5	39.8
		MCDB	47.8	52.7	63.1	67.4	74.1	81.1	85.0	83.2	80.4	69.9	61.9	49.6
	5%	WB	36.6	38.2	44.3	49.2	57.1	63.4	65.5	63.9	59.1	50.7	43.2	37.2
		MCDB	44.8	47.2	58.2	64.4	71.1	80.0	83.1	81.6	78.6	66.3	56.6	45.8
	10%	WB	34.6	35.3	41.9	46.9	54.7	61.4	64.1	62.4	57.2	48.4	40.7	34.8
		MCDB	41.8	42.4	53.3	59.6	67.2	77.9	81.9	80.4	76.4	62.0	51.9	42.6
Mean Daily Temperature Range	5% DB	MDBR	19.5	20.1	22.7	23.5	23.5	27.0	29.5	29.5	27.9	23.8	22.1	19.8
		MCDBR	21.4	24.5	30.7	32.9	30.6	33.8	33.9	33.8	34.6	33.0	29.4	23.1
		MCWBR	13.6	14.3	15.8	16.0	13.8	13.3	10.8	11.2	12.8	15.2	16.0	14.3
	5% WB	MCDBR	21.0	23.3	28.9	30.0	28.2	30.4	29.8	29.4	30.5	29.9	27.9	22.2
		MCWBR	13.8	14.4	15.4	15.4	13.8	13.6	10.9	10.8	12.0	14.8	15.7	14.3
Clear-Sky Solar Irradiance	taub	0.245	0.252	0.270	0.295	0.320	0.328	0.350	0.350	0.312	0.281	0.257	0.243	
	taud	2.581	2.572	2.567	2.486	2.477	2.474	2.435	2.417	2.515	2.581	2.565	2.574	
	Ebn at Noon	291	306	311	308	301	298	289	286	291	290	282	280	
	Edh at Noon	20	24	27	32	34	34	35	34	29	23	20	18	
All-Sky Solar Radiation	RadAvg	582	847	1249	1591	1917	2263	2312	1986	1527	993	649	475	
	RadStd	34	52	107	98	184	164	86	64	103	87	58	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page